Data set link: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

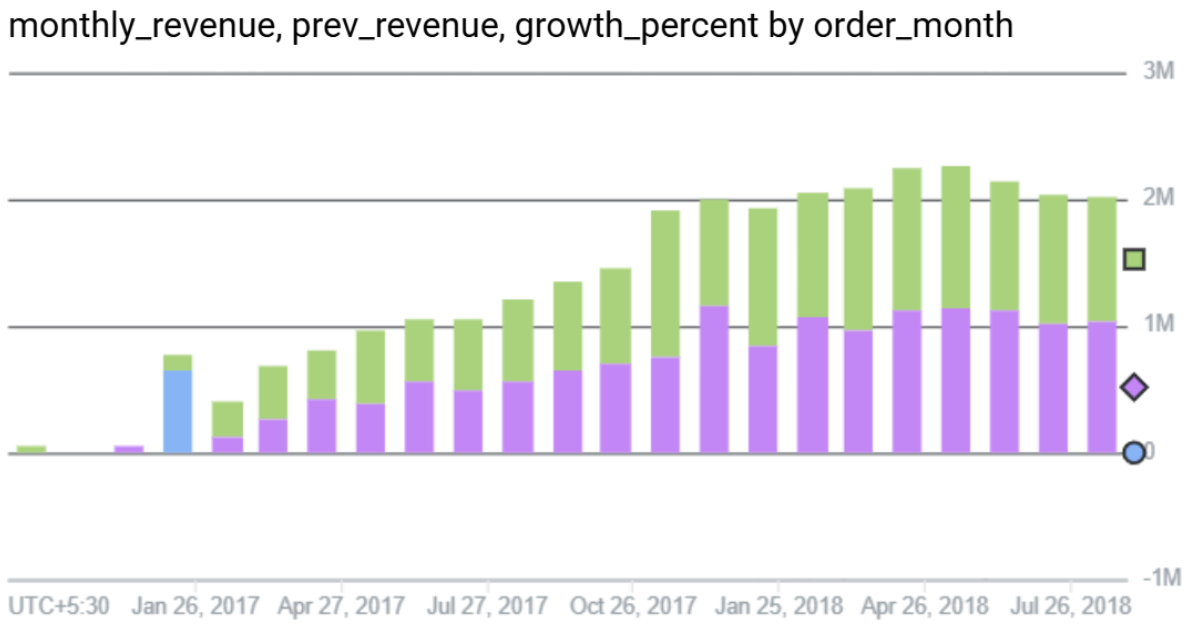
Outputs:

1) **Monthly revenue**
   Sql Query execution:
   https://console.cloud.google.com/bigquery?sq=1012290132513:643be7a1e76e44a0b90b3147b89f28bc

   Output:

Order Month	Monthly Revenue	Previous Revenue	Growth Percent
2016-10-01	$46,566.71	-	-
2016-12-01	$19.62	$46,566.71	-99.96%
2017-01-01	$127,545.67	$19.62	649,979.87%
2017-02-01	$271,298.65	$127,545.67	112.71%
2017-03-01	$414,369.39	$271,298.65	52.74%
2017-04-01	$390,952.18	$414,369.39	-5.65%
2017-05-01	$567,066.73	$390,952.18	45.05%
2017-06-01	$490,225.60	$567,066.73	-13.55%
2017-07-01	$566,403.93	$490,225.60	15.54%
2017-08-01	$646,000.61	$566,403.93	14.05%
2017-09-01	$701,169.99	$646,000.61	8.54%
2017-10-01	$751,140.27	$701,169.99	7.13%
2017-11-01	$1,153,528.05	$751,140.27	53.57%
2017-12-01	$843,199.17	$1,153,528.05	-26.90%
2018-01-01	$1,078,606.86	$843,199.17	27.92%

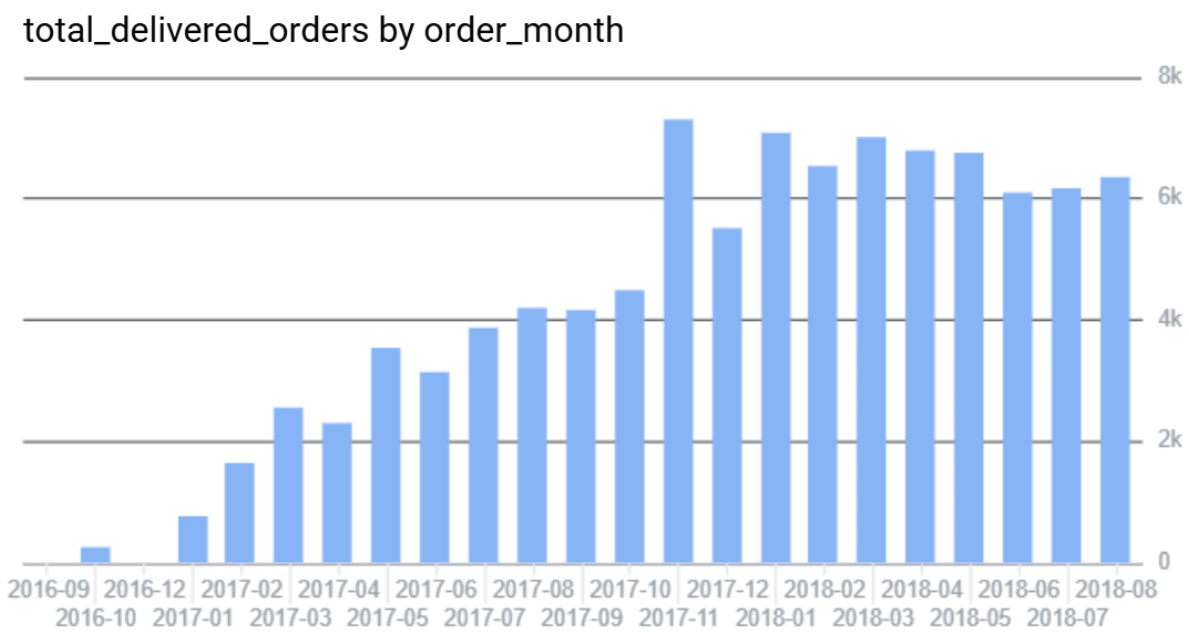
2018-02-01	$966,510.88	$1,078,606.86	-10.39%
2018-03-01	$1,120,678.00	$966,510.88	15.95%
2018-04-01	$1,132,933.95	$1,120,678.00	1.09%
2018-05-01	$1,128,836.69	$1,132,933.95	-0.36%
2018-06-01	$1,012,090.68	$1,128,836.69	-10.34%
2018-07-01	$1,027,903.86	$1,012,090.68	1.56%
2018-08-01	$985,414.28	$1,027,903.86	-4.13%



2) **Monthly order count**
Sql Query link:
https://console.cloud.google.com/bigquery?sq=1012290132513:6976ba3bd7f94e93ad13be5f8111f00d
Output:

order_month	total_delivered_orders
2016-09	1

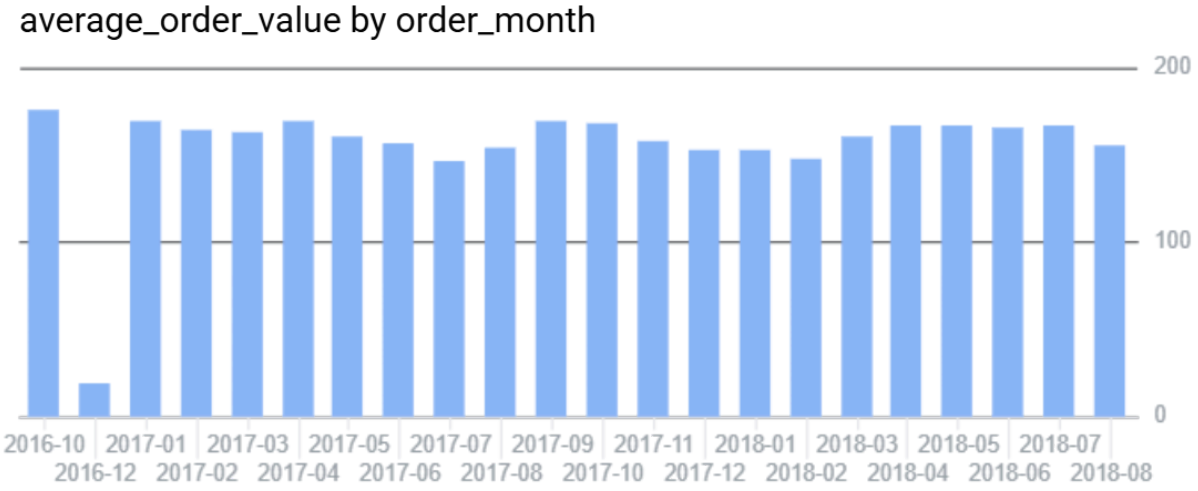
2016-10	265
2016-12	1
2017-01	750
2017-02	1653
2017-03	2546
2017-04	2303
2017-05	3546
2017-06	3135
2017-07	3872
2017-08	4193
2017-09	4150
2017-10	4478
2017-11	7289
2017-12	5513
2018-01	7069
2018-02	6555
2018-03	7003
2018-04	6798
2018-05	6749
2018-06	6099
2018-07	6159
2018-08	6351



3) **Average order value**
   Sql Query:
   https://console.cloud.google.com/bigquery?sq=1012290132513:a6daf24977a94dc0bbc0e65535480438
   Output:

order_month	average_order_value
2016-10	175.72
2016-12	19.62
2017-01	170.06
2017-02	164.13
2017-03	162.75
2017-04	169.76
2017-05	159.92
2017-06	156.37
2017-07	146.28

2017-08	154.07
2017-09	168.96
2017-10	167.74
2017-11	158.26
2017-12	152.95
2018-01	152.58
2018-02	147.45
2018-03	160.03
2018-04	166.66
2018-05	167.26
2018-06	165.94
2018-07	166.89
2018-08	155.16



4) **Top 10 prod by revenue**

Sql Query:
https://console.cloud.google.com/bigquery?sq=1012290132513:97307ada0ed1463d9055c614d40fd9ef

Output:

product_id	total_product_revenue	revenue_rank
bb50f2e236e5eea0100680137654686c	67258.030000000042	1
d1c427060a0f73f6b889a5c7c61f2ac4	58957.310000000085	2
6cdd53843498f92890544667809f1595	57933.729999999952	3
99a4788cb24856965c36a24e339b6058	49907.499999999869	4
3dd2a17168ec895c781a9191c1e95ad7	47876.060000000005	5
d6160fb7873f184099d9bc95e30376af	47314.179999999978	6
aca2eb7d00ea1a7b8ebd4e68314663af	44197.750000000204	7
5f504b3a1c75b73d6151be81eb05bdc9	41725.81	8
25c38557cf793876c5abdd5931f922db	40311.95	9
53b36df67ebb7c41585e8d54d6772e08	39713.489999999911	10

5) **Top 10 by sales**
Sql Queries:
https://console.cloud.google.com/bigquery?sq=1012290132513:20edfecead6a4e0387934360501b0d6a
Output:

product_id	total_orders	sales_rank

99a4788cb24856965c36a24e339b6058	456	1
aca2eb7d00ea1a7b8ebd4e68314663af	425	2
422879e10f46682990de24d770e7f83d	352	3
d1c427060a0f73f6b889a5c7c61f2ac4	313	4
389d119b48cf3043d311335e499d9c6b	309	5
53b36df67ebb7c41585e8d54d6772e08	304	6
368c6c730842d78016ad823897a372db	291	7
53759a2ecddad2bb87a079a1f1519f73	287	8
154e7e31ebfa092203795c972e5804a6	262	9
2b4609f8948be18874494203496bc318	254	10

6) **Repeat vs new customers**
   Sql Query:
   https://console.cloud.google.com/bigquery?sq=1012290132513:a5256aca8fa1479cbbf9ff87e9a2d745
   Output:

new_customers	repeat_customers	repeat_customer_rate_percent
96478	0	0.0

7) **Customer lifeline**

Sql Query:
https://console.cloud.google.com/bigquery?sq=1012290132513:1d1890c33be64e0cb74939744b9075da
Output:

customer_id	total_customer_revenue
1617b1357756262bfa56ab541c47bc16	13664.08
ec5b2ba62e574342386871631fafd3fc	7274.88
c6e2731c5b391845f6800c97401a43a9	6929.31
f48d464a0baaea338cb25f816991ab1f	6922.21
3fd6777bbce08a352fddd04e4a7cc8f6	6726.66
05455dfa7cd02f13d132aa7a6a9729c6	6081.54
df55c14d1476a9a3467f131269c2477f	4950.34
24bbf5fd2f2e1b359ee7de94defc4a15	4764.34
3d979689f636322c62418b6346b1c6d2	4681.78
1afc82cd60e303ef09b4ef9837c9505c	4513.32
cc803a2c412833101651d3f90ca7de24	4445.5
926b6a6fb8b6081e00b335edaf578d35	4194.76
35a413c7ca3c69756cb75867d6311c0d	4175.26
e9b0d0eb3015ef1c9ce6cf5b9dcbee9f	4163.51
3be2c536886b2ea4668eced3a80dd0bb	4042.74
eb7a157e8da9c488cd4ddc48711f1097	4034.44
c6695e3b1e48680db36b487419fb0398	4016.91
31e83c01fce824d0ff786fcd48dad009	3979.55
addc91fdf9c2b3045497b57fc710e820	3826.8
19b32919fa1198aefc0773ee2e46e693	3792.59

8) Cancellation rates

Sql Query:
https://console.cloud.google.com/bigquery?sq=1012290132513:4be4957e6ebb4657a0e5dc225885d65b
Output:

total_orders	cancelled_orders	cancellation_rate_percent
99441	625	0.63

9) cancellation after approval

Sql Query:
https://console.cloud.google.com/bigquery?sq=1012290132513:1019d2d8ffb44e488f50282817ceaffc
Output:

Status	value
cancelled_after_approval	484

10) delivery vs review

Sql Query:
https://console.cloud.google.com/bigquery?sq=1012290132513:b0ae86ee08944d2190e1a5dfec1b8441
Output:

review_score	average_delivery_days
1	20.85
2	16.19
3	13.79
4	11.85
5	10.22

Summary of E-commerce Data Analysis

The provided analysis of the Brazilian E-commerce dataset reveals key performance indicators across several dimensions, including revenue growth, order volume, product performance, customer retention, and operational efficiency (cancellation and delivery times).

Key Performance Indicators (KPIs)

- **Overall Growth and Revenue:** Monthly revenue demonstrated significant volatility, especially in the early months. A major anomaly was observed between October 2016 and December 2016, showing a massive decline followed by an exponential jump in January 2017. Revenue generally trended upward through late 2017, peaking in November 2017 ($1.15 million), but stabilized or slightly declined in 2018, finishing August 2018 at $985,414.28.
- **Order Volume:** The number of orders increased steadily, rising from single-digit counts in 2016 to peak monthly volumes of over 7,000 in January and March 2018. This indicates a growing customer base and/or increasing purchase frequency.
- **Average Order Value (AOV):** AOV generally remained consistent, fluctuating between approximately $146 and $176 over the recorded period (excluding the 2016-12 anomaly). This stability suggests that while the business is growing in volume, the average basket size per transaction is not seeing dramatic shifts.

Product and Customer Analysis

- **Top Products:** There is a notable difference between top products by revenue and top products by sales volume.
    - The top-selling product (product_id: 99a4788cb24856965c36a24e339b6058) ranks 4th in total revenue, suggesting it is a high-volume, lower-priced item.
    - The highest-revenue product (product_id: bb50f2e236e5eea0100680137654686c) ranks outside the top 10 by sales, indicating it is a high-value, lower-volume item.
- **Customer Lifetime Value (CLV):** The top customer generated a total revenue of $13,664.08, highlighting the value of high-spending individual clients.
- **Customer Retention:** The dataset indicates an extremely low repeat customer rate (0.0%), with 96,478 new customers and 0 repeat customers. **This suggests a critical issue with customer retention or a potential data limitation where repeat orders are not correctly linked to original customer IDs.**

Operational Efficiency

- **Cancellation Rate:** The overall cancellation rate is low at **0.63%** (625 cancellations out of 99,441 total orders).
    - A significant portion of these cancellations (484) occurred **after** payment approval, suggesting problems with inventory, fulfillment, or logistics post-sale.

- **Delivery and Review Score Correlation:** There is a clear and strong inverse correlation between review score and delivery time.
    - Orders with the lowest review score (1) had the longest average delivery time (20.85 days).
    - Orders with the highest review score (5) had the shortest average delivery time (10.22 days). **This confirms that timely delivery is a primary driver of customer satisfaction.**